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MILITARY SPECIFICATION

# RELAY, CURRENT, TYPE D-1
# (115 VOLTS, 400 CYCLE - SINGLE POLE)

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

## 1. SCOPE

1.1 This specification covers one type of normally closed, single-pole, 115V, 400-cycle, a-c relay, designated Type D-1.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publication, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-D-5028 | Drawings and Data Lists: Preparation of Manufacturers' (For Production Aircraft, Guided Missiles, Engines, Accessories, and Other Auxiliary Equipment) |
| MIL-E-5272 | Environmental Testing, Aeronautical and Associated Equipment, General Specification for |
| MIL-P-14 | Plastic-Materials, Molding and Plastic-Parts, Molded; Thermosetting |
| MIL-P-3115 | Plastic-Material, Laminated, Thermosetting, Sheets, Paper-Base, Phenolic-Resin |
| MIL-P-6064 | Packaging of Lightweight Aircraft Accessories |
| MIL-S-7742 | Screw Threads, Standard, Aeronautical |

STANDARDS

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U. S. Military Property |

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

| | |
|---|---|
| No. 143 | Specifications and Standards; Use of |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials.-

3.1.1 Protective treatment.- When materials are used in the construction of the relay that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.1.2 Selection of materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.1.2.1 Standard parts.- Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2 Design and construction.- The relay shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.2.1 The relay shall be so constructed that adjustment and repairs can be easily made by the personnel of the operating units and overhaul bases.

3.2.1.1 No special tools shall be required for installation, adjustment, or repair of the relay.

3.2.2 The relay shall be constructed to insure proper operation when mounted in any position, and the contacts and holding coil shall be totally enclosed for mechanical protection against sand and dust.

3.2.3 Mounting bracket.- The relay mounting bracket shall be an integral part of the relay housing, or it shall be securely attached thereto in a manner to prevent any movement between the relay and mounting bracket in service use.

3.2.4 Movable contacts.- The relay movable contacts shall be so secured that when the contact cover is removed the relay will be capable of operating satisfactorily without making any adjustments to the assembly.

3.2.5 Holding coil winding.- The relay holding coil shall consist of not more than one coil, evenly wound, properly insulated, and covered.

3.2.6 Holding coil heads.- Relay holding coil heads, if used, shall be of phenolic material conforming to Specifications MIL-P-14 and MIL-P-3115.

3.2.7 Contact base.- The contact base shall be of phenolic material conforming to Specifications MIL-P-14 and MIL-P-3115.

3.2.8 Contact alinement.- The movable contacts shall be self-alining with the stationary contacts and shall be positive in operation.



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3.2.9 Rating.- The rating of the relay contacts shall be not less than 3 amp at 29V dc.

3.2.10 Terminal posts.-

3.2.10.1 Both leads of the relay holding coil shall be brought out on one end of the relay to terminal posts and insulated from the coil housing. The leads shall not be exposed to possible damage in service. The load terminal posts shall be attached to the opposite end of the relay and insulated from the coil housing.

3.2.10.2 The coil and load terminal posts shall be threaded with No. 6-32 NC-2 threads, and shall be provided with the proper nuts and lock washers or locknuts for attaching the cable terminals, and shall be protected against damage in service.

3.3 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.4 Screw threads.- Screw threads shall be in accordance with Specification MIL-S-7742.

3.5 Weight.- The weight of the relay shall be not more than 0.6 pound.

3.6 Finishes and protective coatings.-

3.6.1 Metal parts.- All exposed steel parts shall be corrosion-resistant steel or protected by a plating of corrosion-resistant metal.

3.6.2 Phenolic.- Phenolic or any other material used in the construction of the relay shall be suitably treated or protected against moisture absorption.

3.7 Performance.- Unless otherwise specified, the relay shall be capable of operating satisfactorily under the following conditions:

(a) Temperature: Temperatures ranging from -54° to +71°C.

(b) Humidity: Relative humidity up to 100 percent including conditions wherein condensation takes place in the form of both water and frost.

(c) Altitude: Pressures ranging from 30 inches Hg down to 3.4 inches Hg (approximately an altitude of 50,000 feet).

(d) Salt spray: Exposure to salt sea atmosphere.

(e) Vibration: Vibration incident to service use.

(f) Sand and dust: Sand and dust particles as will be encountered in desert areas.

(g) Acceleration and shock: Acceleration and shock forces as anticipated.

3.8 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.9 Workmanship.-

3.9.1 General.- The relay, including all accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, and riveting; alinement of parts and tightness of assembly screws, and bolts; etc.



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3.9.2 Riveting.- Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.9.3 Cleaning.- The relay shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of relays shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests performed on samples representative of the production of the item after the award of contract, to determine that the production meets the requirements of this specification.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 Test conditions.-

4.2.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3 Preproduction tests.-

4.3.1 Sampling instructions.- One relay as specified in the contract will be tested for design approval by the procuring activity, or when so specified in the contract, at the contractor's plant under the supervision of the procuring activity.

4.3.2 Tests.- The Preproduction tests of relays shall consist of all the tests described under Acceptance tests and, in addition, the following tests.

4.3.2.1 Altitude operation.- The relay shall be subjected to 10,000 continuous cycles in accordance with the Resistive load test specified in paragraph 4.3.2.2 at an altitude of 50,000 ±2,000 feet without welding of the contacts.

4.3.2.2 Resistive load.- The relay shall be subjected to test for making and breaking a 3-amp resistive load at 29.0V dc for 40,000 operations with 115V, 400-cycle ac across the coil. The operations shall be conducted at the rate of 60 cpm with an approximate 0.5-second "on" and "off" period.

4.3.2.3 The following tests shall be conducted in accordance with the specified procedures of Specification MIL-E-5272.

4.3.2.3.1 Vibration resistance: Procedure I.- A test lamp or other suitable indicating device shall be connected across the main contacts while the relay is undergoing this test to determine that the vibration does not open the contacts with no voltage across the coil or close them with 85.0V across the coil. The relay contacts shall satisfactorily open during this test with 85.0V across the coil and stay open as low as 70.0V.



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4.3.2.3.2 Acceleration resistance: Procedure II.- The relay shall be subjected to acceleration forces of 10g. The time required to attain the 10g acceleration shall be not more than 2 seconds. A test lamp or a suitable indicating device shall be connected across the main contact terminals while the relay is undergoing this test, to determine that the acceleration does not result in closing or opening of the contacts.

4.3.2.3.3 Low temperature: Procedure II.- At the end of this period and with the relay at the specified temperature, the relay shall be subjected to the test specified in paragraph 4.3.2.3.9. The current consumed shall not exceed that specified. The relay shall then be tested intermittently for closing and opening voltages until it attains a temperature of 20° to 30°C. The voltage characteristics during this test shall conform to the applicable test specified in paragraph 4.4.1.4.

4.3.2.3.4 High temperature: Procedure I.- At the completion of this period and with the relay at the specified temperature, the relay shall be subjected to the applicable test specified in paragraph 4.3.2.3.9. The relay shall then be tested intermittently for closing and opening voltages until it attains a temperature of 20° to 30°C. The voltage characteristics during this test shall conform to the applicable test specified in paragraph 4.4.1.4.

4.3.2.3.5 Humidity: Procedure I for a period of 30 days.- During this test, a potential of 150V rms shall be applied to the terminals and other exposed metal parts. There shall be no insulation failure as a result of this test. Immediately after this test, the relay shall be subjected to the applicable Operating voltage test specified in paragraph 4.4.1.4, and then shall be subjected to 10,000 continuous cycles in accordance with the test specified in paragraph 4.3.2.2.

4.3.2.3.6 Sand and dust: Procedure I.-

4.3.2.3.7 Salt spray.- The duration of the test period shall be 50 hours. At the completion of this test, the relay shall satisfactorily pass the operating voltage characteristics in accordance with the tests specified herein and shall be tested for operation at 10,000 continuous cycles as specified in paragraph 4.3.2.2, without welding of the contacts.

4.3.2.3.8 Continuous operation.- The relay shall be subjected to 5 hours continuous operation with 115V applied to the holding coil, and at the completion of this period, the main contacts shall remain open on 70V when the applied voltage is reduced from 115V.

4.3.2.3.9 Current consumption.- The relay shall be tested to determine that the holding coil consumes no more than 0.06 amp when operated on 115V ac.

4.4 Acceptance tests.- Acceptance tests shall consist of Individual tests and Sampling tests.

4.4.1 Individual tests.- Each relay shall be subjected to the following tests. In addition, each relay shall be subjected to any other test specified herein which the Inspector considers necessary to determine compliance with the requirements of this specification.

4.4.1.1 Examination of product.- Each relay shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.1.2 Insulation test.- The relay shall be tested with an alternating-current potential of 1,250V rms at commercial frequency between each terminal and other exposed metal parts for 60 seconds. There shall be no failure as a result of this test.



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4.4.1.3 Opening voltage.- The relay shall be subjected to test to determine that the main contacts open with an alternating-current potential of 85.0V ac, or less, applied to the holding coil terminals and to determine that the main contacts remain open on 70V when the opening voltage applied to the holding coil is reduced from 115V.

4.4.1.4 Operating voltage.- The relay shall be designed to withstand a potential of 130V ac applied to the holding coil terminals for a 3-minute period and at the completion of the test shall be subjected to the test specified in paragraph 4.4.1.2.

4.4.2 Sampling tests.- One relay which has been manufactured under essentially the same conditions shall be selected at random from each lot of 500 or fraction thereof on the order and subjected to the following tests.

4.4.2.1 The sample relay shall be subjected to the Vibration resistance test specified in paragraph 4.3.2.3.1.

4.4.2.2 The sample relay shall be subjected to the Acceleration resistance test specified in paragraph 4.3.2.3.2.

4.4.3 Rejection and retest.- When Sampling tests are specified on a number of relays that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, acceptance of all relays shall be withheld until the extent and cause of failure is determined. For operational reasons, Individual tests may be continued pending investigation of a Sampling test failure, but the final acceptance of the relay is contingent upon the Inspector's decision regarding the over-all conformance of the relay to specification requirements. If investigation indicates that the defects may exist on relays previously accepted, full particulars concerning the defects found, including recommendations for correction, shall be furnished to the procuring activity.

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with this specification, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

## 5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and packing.- Unless otherwise specified, each relay shall be packaged and packed for shipment in accordance with Specification MIL-P-6064 and the following.

5.2.1 Each relay shall be wrapped in grade A paper. Four wrapped relays shall be packed as a unit in a type II container.

5.3 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Relay, Current, (115V, 400 Cycle - Single Pole), Type D-1, Specification MIL-R-8373A, Manufacturer's Part No. *, Federal Stock No. * (if no FSN available, leave space therefor).

*Applicable data to be entered by the contractor.



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6. NOTES

6.1 Intended use.- The Type D-1 relay covered by this specification is intended for use in aircraft as a means of remotely controlling the operation of an indicating light for indication of inverter failure.

6.2 Ordering data.- Invitations for bids, contracts, and purchase orders should state the conditions for the following.

6.2.1 Preproduction tests.- It is expected that the contract or purchase order will specify that one relay will be required as a preproduction sample and that the preproduction sample will be subjected to the Preproduction tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

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